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# SemCSE: Semantic Contrastive Sentence Embeddings Using LLM-Generated Summaries For Scientific Abstracts

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# Overview

- **Problem**: Embedding models for scientific abstracts **do not focus on semantic content**!
- Current embedding models often use citation links as indicator for similarity, and train embedding models to embed papers that cite each other to similar locations.
- But: **Citation links do not reliably indicate semantic similarity**! Instead, citations often link to vaguely related foundational work or own work, or only reference very specific aspects that are not included in the abstract.
- We therefore aim at creating an embedding model that **focuses on the actual semantic content**, thus likely improving embedding quality and downstream performance.

# Approach

- We propose to use a dataset of highly semantically related scientific texts, which can be used to train the embedding model to embed those to nearby locations in the embedding space

- How to acquire such a dataset?
  → Solution: Leverage LLMs

# Dataset Creation

- We use a dataset consisting of 350K scientific abstracts from various domains, sampled from the datasets in the SciRepEval benchmark
- We then use Llama-3-8B in its pure autoregressive language model version
- We append strings like: “In summary, our research is concerned with” to the abstract and let the LLM complete this sentence
  → This method creates summaries in the exact same style as the abstract, thus approximately preserving the data distribution of scientific abstract sentences

# Model Training

- The model is trained via a triplet loss:

$$\mathcal{L}(e_{\mathrm{a}}, e_{+}, e_{-}) = \mathrm{relu}(d(e_a, e_+) - d(e_a, e_-) + 1)$$

- Given two related summaries for the same paper, it defines one of them as the anchor and one as the positive
- We then sample a random negative (i.e., a summary for a different paper)
- The loss forces the embeddings of the related summaries to be at least 1 unit closer than the distance between anchor and negative

→ Semantically related texts are placed in proximity, unrelated ones further apart!

# Generalization to longer texts

- Note that the model is trained solely on summaries

- To evaluate generalization to longer texts, we create a retrieval task:

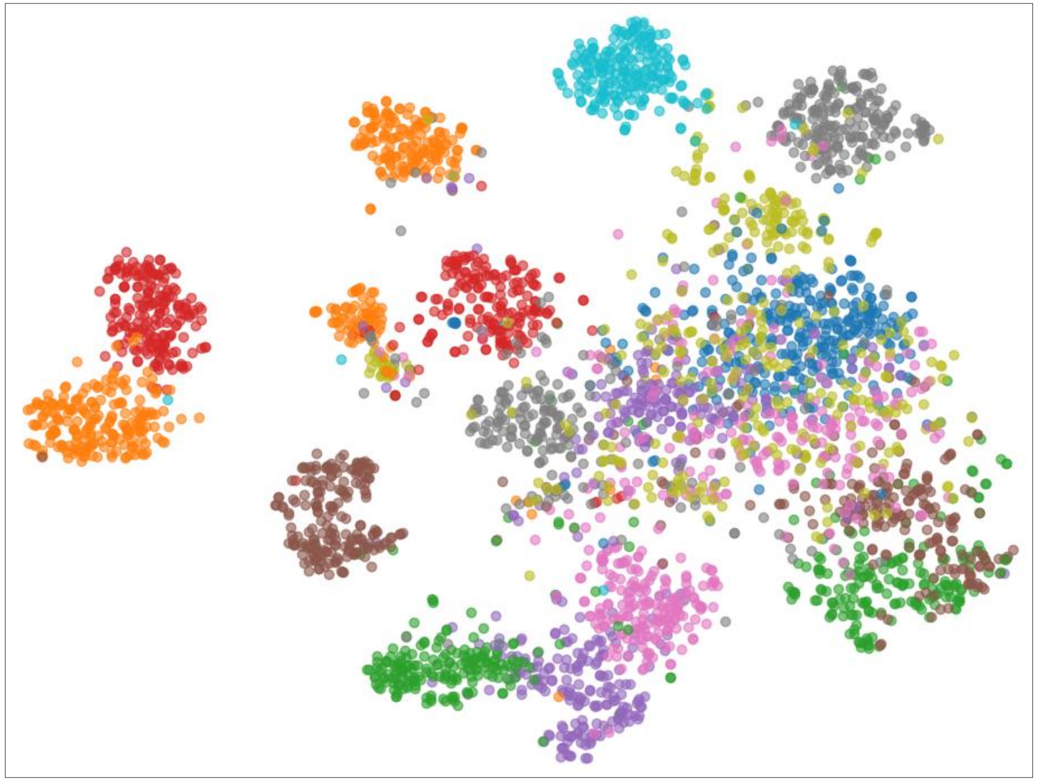
  → We embed a scientific abstract and a corresponding summary
  → The model must identify the correct summary out of a set of 900 summaries
  → The rank at which the correct summary is retrieved is the evaluation score

- Over all samples, the correct match was on average retrieved at rank 1.542, thus showing that the embeddings of the longer abstracts are also meaningful

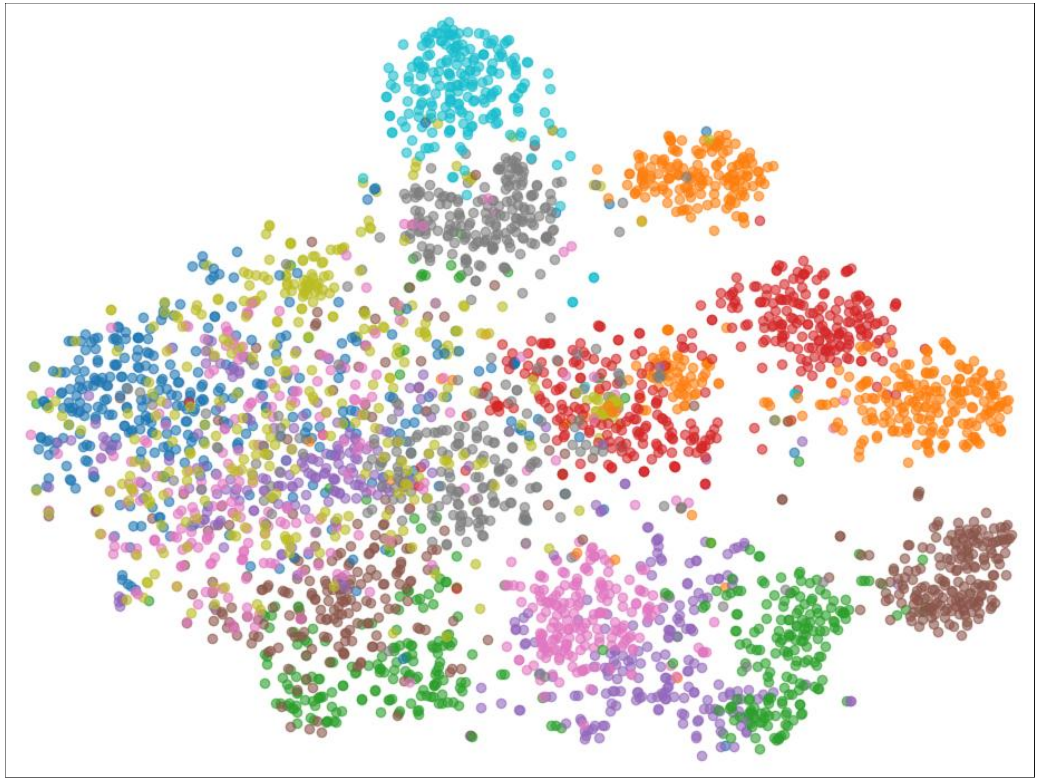
# Evaluation: SciRepEval

- We evaluate SemCSE on the SciRepEval benchmark
- It comprises 24 different tasks to evaluate scientific text embeddings

| **Model** | Parameters | Classification | Regression | Proximity | Search | Average |
|---|---|---|---|---|---|---|
| SciBERT | 109M | 63.86 | 27.34 | 66.25 | 68.19 | 57.42 |
| SciDeBERTa | 183M | 60.99 | 27.00 | 62.74 | 67.83 | 55.18 |
| SPECTER | 109M | 67.73 | 25.37 | 80.05 | 74.89 | 64.28 |
| SciNCL | 109M | <u>68.04</u> | 25.22 | <u>81.18</u> | 77.32 | 65.08 |
| SPECTER2 base | 109 | 66.95 | <u>27.75</u> | 81.10 | 78.42 | 65.46 |
| SPECTER2 proximity | 110 | 66.37 | 26.85 | **81.41** | 77.75 | 65.15 |
| all-MiniLM-L6-v2 | 22M | 64.04 | 20.06 | 80.74 | 79.63 | 63.05 |
| jina-v2 | 137M | 63.99 | 23.76 | 80.11 | 80.40 | 63.69 |
| jina-v3 | 572M | 65.66 | 24.84 | 79.98 | <u>80.60</u> | 64.34 |
| RoBERTa SimCSE | 355M | 67.16 | 22.95 | 75.51 | 76.97 | 62.10 |
| NvEmbed-V2 | 7.9B | 65.62 | **29.94** | 81.16 | **82.84** | **66.19** |
| SemCSE (Ours) | 183M | **69.52** | 27.58 | 80.21 | 78.56 | <u>65.76</u> |



**SemCSE**



**SPECTER2 base**

# SciRepEval issues

- The SciRepEval benchmark can not be used to reliably evaluate the ability to produce semantically accurate embeddings
- Reason: Many tasks do not rely on semantic relatedness

  → Multiple tasks for same author detection
  → Citation Link Prediction
  → Peer Review Score prediction
  → Tweet Mention Prediction

# Solution: New Semantic Benchmark

- We create a new benchmark for evaluating semantic embedding capabilities

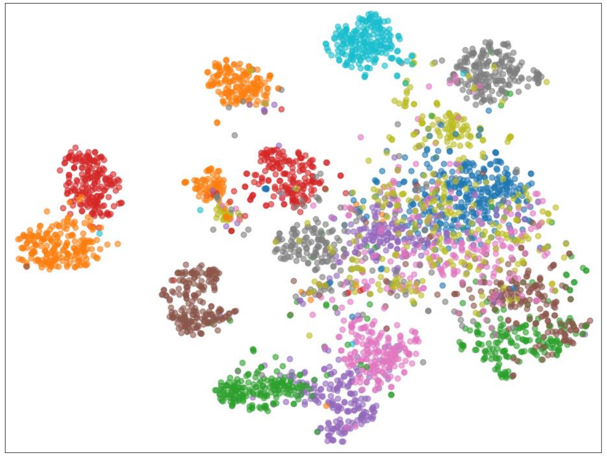
- Three tasks require the matching of different representations of the same study
  → Matching a paper title to its abstract
  → Matching two halves of the same abstract
  → Matching a LLM-generated query to the corresponding abstract

- The fourth tasks evaluates the clustering of the SciDocs MAG dataset:
  → Samples with the same label should be clustered together

| **Model** | Parameters | Title-Abstr. ↓ | Abstr.-Segments ↓ | Query ↓ | Clustering ↑ | Perf. ↑ |
|---|---|---|---|---|---|---|
| SciBERT | 109M | 807.74 | 214.37 | 213.45 | 0.569 | 0.000 |
| SciDeBERTa | 183M | 1479.09 | 861.55 | 2465.26 | 0.460 | 0.000 |
| SPECTER | 109M | 10.25 | 12.23 | 2.18 | 0.692 | 0.119 |
| SciNCL | 109M | 5.68 | 7.35 | 2.29 | 0.702 | 0.357 |
| SPECTER2 base | 109M | 4.52 | 5.10 | 1.17 | 0.666 | 0.553 |
| SPECTER2 proximity | 110M | 5.34 | 5.80 | 1.46 | 0.666 | 0.395 |
| all-MiniLM-L6-v2 | 22M | <u>3.09</u> | 8.19 | 1.11 | <u>0.730</u> | 0.771 |
| Jina-v2 | 137M | 3.29 | 8.77 | 1.29 | 0.703 | 0.600 |
| Jina-v3 | 572M | 3.45 | 6.96 | **1.01** | 0.719 | 0.783 |
| RoBERTa SimCSE | 355M | 23.71 | 44.24 | 8.92 | 0.696 | 0.116 |
| NvEmbed-V2 | 7.9B | 3.38 | <u>3.84</u> | <u>1.02</u> | 0.721 | <u>0.866</u> |
| SemCSE (Ours) | 183M | **2.47** | **2.68** | 1.23 | **0.739** | **0.925** |

# Analysis: Are different summaries needed?

- The popular SimCSE approach used the same sentence twice for embedding model training
  → We instead use two different summaries of the same abstract



Different Summaries

Same Input

| Model | Dataset | Clf. | Regr. | Prox. | Search | SRE Avg. | Title-Abstr. | Abstr.-Segments | Query | Clustering |
|---|---|---|---|---|---|---|---|---|---|---|
| Full | 350K | 69.52 | 27.58 | 80.21 | 78.56 | 65.76 | 2.47 | 2.68 | 1.23 | 0.739 |
| Full | 175K | 69.26 | 27.19 | 80.36 | 78.54 | 65.67 | 3.40 | 3.21 | 1.29 | 0.732 |
| Full | 87.5K | 69.01 | 27.39 | 80.35 | 78.58 | 65.65 | 3.35 | 2.63 | 1.24 | 0.731 |
| Full | 35K | 69.32 | 27.18 | 80.18 | 77.81 | 65.51 | 3.65 | 3.24 | 1.24 | 0.734 |
| Full | 17.5K | 68.31 | 27.00 | 80.04 | 78.37 | 65.23 | 3.76 | 3.47 | 1.45 | 0.723 |
| Full | 8.75K | 68.95 | 27.05 | 79.96 | 76.53 | 65.14 | 4.34 | 3.73 | 1.99 | 0.725 |
| Full | 3.5K | 68.36 | 27.07 | 79.94 | 76.73 | 65.01 | 5.93 | 4.04 | 2.18 | 0.728 |
| Just Summaries | 350K | 68.78 | 26.89 | 80.21 | 77.51 | 65.28 | 6.04 | 3.52 | 2.11 | 0.732 |
| Same Input | 350K | 66.59 | 27.23 | 74.29 | 69.94 | 61.48 | 180.01 | 59.86 | 251.46 | 0.645 |
| Cosine Similarity | 350K | 69.76 | 25.88 | 80.63 | 79.33 | 65.72 | 2.84 | 3.17 | 1.13 | 0.735 |

- Different summaries are required for good performance!
- Original SimCSE used dropout for data augmentation, so that embeddings of the two equal sentences are different
- Out method can be seen as an extension that uses are more complex data augmentation strategy for creation more strongly differing inputs that are still related